



**POLITECNICO
MILANO 1863**

**DIPARTIMENTO DI ELETTRONICA
INFORMAZIONE E BIOINGEGNERIA**

Data and Information Quality Project Report

The Metropolitan Museum of Art Dataset
Assigned Dataset: 4

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Academic Year 2025–2026

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1 Setup Choices

1.1 Libraries

The work was performed in a Python environment (version 3.11.3). The experimental setup includes a wide variety of libraries that are commonly used in the data science domain. The list of libraries, together with a brief motivation for their use, is reported below:

- **pandas** (version 2.2.3): allows storing the dataset in a `DataFrame` object, which models data as a relation of a structured database and provides useful methods to access and modify values in a simple and efficient way.
- **recordlinkage** (version 0.16): provides the most important methods required for the duplicate detection phase.
- **scikit-learn** (version 1.6.0) and **mlxtend** (version 0.23.3): include ready-to-use feature extraction techniques, machine learning models, and data mining algorithms.
- **Desbordante**(version 2.3.2) is an open-source framework for discovering data dependencies and profiling data quality in large datasets. It provides scalable, extensible algorithms for tasks such as functional dependency and constraint discovery.
- **numpy**(version 1.26.4) scientific computing library used for linear algebra and statistical analysis.
- **matplotlib** (version 3.9.4) and **seaborn** (version 0.13.2): libraries used for data visualization.
- **missingno**(version 0.5.2) is a Python library used to *visualize missing (NaN) values* in datasets. It provides plots such as matrix, bar, heatmap, and dendrogram to quickly understand missing-data patterns.
- **pyECLAT**(version 1.0.2) is a Python library for *association rule mining* using the ECLAT algorithm, efficiently discovering frequent itemsets in transactional datasets.
- **DiffLib** is a Python module for *comparing sequences*, measuring similarity between strings or lists based on matching subsequences.

1.2 Introduction of Dataset

The Metropolitan Museum of Art presents over 5,000 years of art from around the world, providing access and engagement to a global audience through its galleries, exhibitions, public programs, and online platforms. Located at two iconic sites in New York City, The Met Fifth Avenue and The Met Cloisters, the Museum has, since its founding in 1870, sought to be more than a repository of rare objects by contextualizing artworks to reveal new perspectives and cross-cultural connections.

The Metropolitan Museum of Art offers curated datasets describing more than 470,000 artworks from its collection for unrestricted commercial and non-commercial use. These

datasets are released under the Creative Commons Zero (CC0) license, waiving copyright and related rights to the extent permitted by law. The dataset is organized in tabular format, where each row represents an individual artwork and each column corresponds to attributes such as identification, classification, cultural and historical context, geographic origin, legal status, and metadata. It contains categorical, numerical, and binary variables, with missing values (NaN) reflecting unavailable, unknown, or inapplicable information due to the heterogeneous nature of the collection.

1.3 Column Descriptions

1.3.1 Identification and Administrative Information

- **Object Number:** The official accession number assigned by the museum to uniquely identify an artwork within the collection.
- **Object ID:** A unique numerical identifier used internally for database management and referencing purposes.
- **Department:** The curatorial department responsible for the artwork, reflecting the organizational structure of the museum.
- **Repository:** The institution that holds and manages the artwork, which in this dataset corresponds to the Metropolitan Museum of Art.
- **Metadata Date:** The date and time indicating the most recent update of the metadata record.

1.3.2 Visibility and Legal Status

- **Is Highlight:** A boolean indicator specifying whether the artwork is considered a highlight of the collection and is typically featured in exhibitions or curated selections.
- **Is Public Domain:** A boolean flag indicating whether the artwork belongs to the public domain and can be reused without copyright restrictions.
- **Rights and Reproduction:** A textual field providing information about copyright ownership, reproduction permissions, and usage conditions, when applicable.

1.3.3 Object Description and Physical Characteristics

- **Object Name:** A general classification describing the type of object (e.g., painting, sculpture, coin).
- **Title:** The official or commonly used title of the artwork.
- **Medium:** A description of the materials and techniques used to create the artwork.
- **Dimensions:** The physical measurements of the artwork, such as height, width, and depth.
- **Classification:** A higher-level categorization of the artwork, often related to material, function, or artistic tradition.

- **Credit Line:** Information describing how the artwork was acquired, including donors, purchases, or bequests.

1.3.4 Cultural and Historical Context

- **Culture:** The cultural group, civilization, or society associated with the artwork.
- **Period:** The historical period or stylistic era to which the artwork belongs.
- **Dynasty:** The ruling dynasty associated with the artwork, when applicable.
- **Reign:** The specific reign or ruling period linked to the artwork, if relevant.
- **Object Date:** A human-readable description of the date or time span associated with the artwork.
- **Object Begin Date:** The earliest estimated year associated with the creation of the artwork.
- **Object End Date:** The latest estimated year associated with the creation of the artwork.

1.3.5 Artist-Related Information

- **Portfolio:** A thematic or curated grouping to which the artwork belongs.
- **Artist Role:** The role performed by the artist in the creation of the artwork (e.g., painter, sculptor, engraver).
- **Artist Prefix:** Honorifics or prefixes associated with the artist's name.
- **Artist Display Name:** The standardized name of the artist as presented by the museum.
- **Artist Display Bio:** A brief biographical description of the artist.
- **Artist Suffix:** Suffixes or titles associated with the artist's name.
- **Artist Alpha Sort:** A normalized version of the artist's name used for alphabetical sorting.
- **Artist Nationality:** The nationality or cultural affiliation of the artist.
- **Artist Begin Date:** The year marking the beginning of the artist's active or life period.
- **Artist End Date:** The year marking the end of the artist's active or life period.

1.3.6 Geographical Provenance

- **Geography Type:** The type of geographic reference associated with the artwork (e.g., place of creation or discovery).
- **City:** The city associated with the artwork’s origin or discovery.
- **State:** The state or administrative region associated with the artwork.
- **County:** The county or local administrative division related to the artwork.
- **Country:** The country associated with the artwork’s geographic context.
- **Region:** A broader geographic region associated with the artwork.
- **Subregion:** A more specific subdivision within a geographic region.
- **Locale:** A localized geographic reference, such as a town or settlement.
- **Locus:** The specific archaeological context or find location.
- **Excavation:** Information regarding the excavation through which the artwork was recovered.
- **River:** A river associated with the artwork’s geographic or archaeological context, when relevant.

1.3.7 Digital Access and Metadata

- **Link Resource:** A URL linking to the official online museum record for the artwork, providing access to additional metadata and images.

1.4 Project Outline

The project is executed through a structured data preparation pipeline divided into two primary phases: Data Exploration and Data Cleaning. The exploration phase aims to decode the dataset’s semantic structure and identify quality deficits. This involves extensive data profiling ranging from single-column analysis to the discovery of functional and inclusion dependencies to map relationships between attributes. The findings from this stage are used to compute baseline metrics for standard quality dimensions such as completeness, uniqueness, and consistency, effectively diagnosing the “health” of the raw data before any modification occurs.

The second phase, Data Cleaning, implements a corrective workflow designed to address the specific issues identified during profiling. This comprehensive process includes data wrangling steps such as dimensionality reduction to remove administrative noise, regex-based feature extraction to recover temporal data, and column-by-column standardization. The pipeline then advances to complex validation tasks, employing logic-based inconsistency checks, statistical and distance-based outlier detection, and fuzzy deduplication to ensure historical accuracy.

The project concludes with a Performance Analysis, utilizing dimensionality reduction techniques (PCA and t-SNE) to visually and quantitatively verify the improvement in data quality.

2 Pipeline Implementation

2.1 Data Profiling

Data profiling focused on the collection of metadata providing valuable information to guide the subsequent data cleaning steps. In particular, it included the following activities:

- **Basic profiling operations**
- **Single-column analysis**, including:
 - cardinalities;
 - value distributions.
- **Dependency discovery**, including:
 - Functional Dependency;
 - Inclusion Dependency;
 - Matching Dependency;
 - Association Rules.

It is important to note that, despite the availability of automatic profiling tools such as `ydata-profiling`, manual profiling operations were preferred. This choice allowed for greater flexibility in metadata collection, customization of the profiling process according to specific needs, and a deeper exploration of the dataset contents, fostering a better understanding of the data for the subsequent phases of the work.

2.1.1 Basic Data Profiling Operation

The dataset shape and column data types were initially inspected. The original dataset consisted of 448203 rows and 43 columns. Among these, 38 columns had `object` data types (i.e., string-based attributes), while the remaining 3 columns contained `int64` and 2 columns are boolean. The variables can be classified as follows:

- **Objects data:**
 - Object Number;
 - Department;
 - Object Name;
 - Title;
 - Culture;
 - Period;
 - Dynasty;
 - Reign;
 - Portfolio;
 - Artist Role;

- Artist Prefix;
- Artist Display Name;
- Artist Display Bio;
- Artist Suffix;
- Artist Alpha Sort;
- Artist Nationality;
- Artist Begin Date;
- Artist End Date;
- Object Date;
- Medium;
- Dimensions;
- Credit Line;
- Geography Type;
- City;
- State;
- County;
- Country;
- Region;
- Subregion;
- Locale;
- Locus;
- Excavation;
- River;
- Classification;
- Rights and Reproduction;
- Link Resource;
- Metadata Date;
- Repository;

- **Int64 Data:**

- Object Begin Date;
- Object End Date;
- Object ID;

- **Boolean Data:**

- Is Highlight;
- Is Public Domain;

2.1.2 Single Column Analysis

Single-column analysis was aimed at computing attribute cardinalities, the number of unique values, and the number of missing values for each column.

Cardinalities For each column, the number of missing values, non-missing values, and unique values was computed. The results are reported in Table 1.

Number of unique value for each column Since most of the attributes are of type `object`, numerical distribution analysis could not be performed. Instead, the number of unique values for each column was examined. You can see the number of unique values per column in Figure 1.

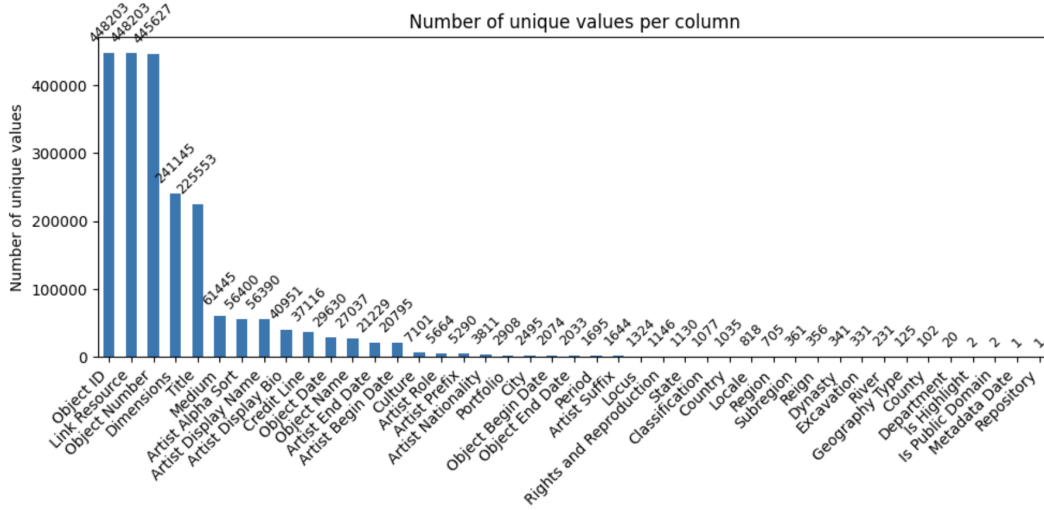


Figure 1: Number of unique values for each column

As shown in Figure 1, the `Object ID`, `Object Number`, and `Link Resource` columns exhibit a significantly higher number of unique values compared to the other attributes that we will discuss about them in other sections.

Number of Missing Values for each Column As shown in Figure 2, more than half of the columns exhibit a high number of missing values.

2.1.3 Dependency Discovery

Its goal is to uncover structural or statistical dependencies that describe how attributes are related to each other, which can be exploited to understand the data better and to support data cleaning and validation. For dependency discovery, the attributes were grouped according to their semantic meaning.

Functional Dependency It is a relationship between two sets of attributes in a dataset, where the value of one attribute (or a set of attributes) uniquely determines the value of another attribute. We grouped the most relevant columns to determine the functional dependency were grouped according to their semantic meaning in order to analyze dependencies within homogeneous sets of features.

Column Name	Missing	Actual	Cardinality
Object Number	0	448203	445626
Is Highlight	0	448203	2
Is Public Domain	0	448203	2
Object ID	0	448203	448203
Department	0	448203	20
Object Name	2635	445568	25986
Title	31447	416756	223338
Culture	261685	186518	7017
Period	376321	71882	1633
Dynasty	425185	23018	329
Reign	437386	10817	328
Portfolio	427833	20370	2812
Artist Role	188294	259909	5664
Artist Prefix	359275	88928	5110
Artist Display Name	187092	261111	56384
Artist Display Bio	224139	224064	40935
Artist Suffix	437991	10212	1621
Artist Alpha Sort	187115	261088	56382
Artist Nationality	252071	196132	3797
Artist Begin Date	232969	215234	20795
Artist End Date	235378	212825	21229
Object Date	15594	432609	29321
Object Begin Date	0	448203	2074
Object End Date	0	448203	2033
Medium	8048	440155	58715
Dimensions	62843	385360	237711
Credit Line	670	447533	36203
Geography Type	389740	58463	125
City	417683	30520	2482
State	439843	8360	1107
County	445715	2488	101
Country	373753	74450	1015
Region	417125	31078	685
Subregion	426487	21716	356
Locale	433108	15095	781
Locus	441264	6939	1288
Excavation	432684	15519	326
River	446100	2103	227
Classification	58279	389924	1077
Rights and Reproduction	425228	22975	1131
Link Resource	0	448203	448203
Metadata Date	0	448203	1
Repository	0	448203	1

Table 1: Summary statistics for dataset columns

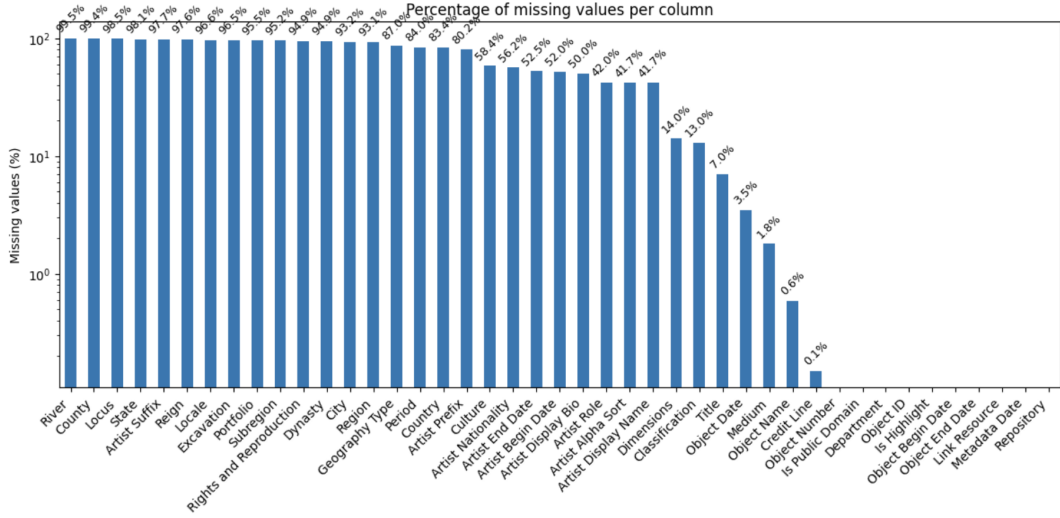


Figure 2: Percentage of missing values for each column

- **Cultural attributes:** this group includes information related to the cultural and historical context of the objects, such as culture of origin, historical period, dynasty, reign, portfolio, classification, and medium.
- **Artist-related attributes :** this set contains information describing the artist associated with each object, including the artist's name, nationality, role, and lifespan, represented by begin and end dates.
- **Geographical attributes:** this group comprises location-based information describing the geographical origin or context of the objects, including city, state, county, country, region, and subregion.

Functional dependencies were analyzed using the *FdMine* algorithm, which discovers functional dependencies by efficiently exploring attribute combinations and identifying deterministic relationships between attributes in the dataset.

```
In [ ]: def run_fd_discovery(df, columns, title):
        print(f"\n=== {title} ===")

        subset = df[columns].dropna(how="all")

        algo = db.fد.algorithms.FdMine()
        algo.load_data(table=subset)
        algo.execute()

        for fd in algo.get_fds():
            print(fd)

In [ ]: run_fd_discovery(df, cultural_cols, "Cultural FDs")
        run_fd_discovery(df, artist_cols, "Artist FDs")
        run_fd_discovery(df, geo_cols, "Geography FDs")

=== Cultural FDs ===

=== Artist FDs ===

=== Geography FDs ===
```

Figure 3: Functional Dependency

As shown in Figure 3, no functional dependencies were identified among the dataset attributes.

Inclusion Dependency Inclusion dependency analysis was performed to verify whether the set of values of one attribute is contained within the set of values of another attribute. This allows the identification of hierarchical or referential relationships among columns. In particular, an approximate inclusion dependency was identified between geographical attributes, indicating that almost all values of **County** are included in the set of values of **State**. The high coverage value (0.98) suggests a strong dependency, with only a small fraction of values violating the containment relationship, likely due to noise or data quality issues.

```
In [ ]: geo_cols = [
        "City",
        "State",
        "County",
        "Country",
        "Region",
        "Subregion"
    ]

    geo_ind = discover_inclusion_dependencies(
        df_clean,
        geo_cols,
        min_coverage=0.8 # allow small noise
    )

    geo_ind
```

Out []:	A ⊆ B	coverage
0	County ⊆ State	0.98

Figure 4: Inclusion Dependency

Matching Dependency Matching dependency analysis was conducted to identify similarity-based relationships between pairs of records. The objective was to verify whether records that are similar with respect to a given attribute also exhibit consistent values on another attribute.

The analysis evaluates approximate similarities between attribute values using string-based comparison measures. When the similarity between two records exceeds a pre-defined threshold, the corresponding values of another attribute are expected to match. Only dependencies supported by a sufficient number of record pairs and exhibiting high confidence are retained.

This approach enables the discovery of approximate matching dependencies, which are particularly useful in the presence of noisy or inconsistent textual data, and supports entity resolution and data deduplication tasks.

As shown in Figure 5, object-related attributes exhibit strong matching dependencies between **Title** and **Object Name**, as well as between **Medium** and **Classification**, indicating internal consistency in object descriptions.

Figure 6 reports the results for geographical attributes, where high-confidence matching dependencies confirm the hierarchical structure of location data, particularly between **Subregion**, **Region**, and **Country**.

Out[]:		MD	support	confidence
0	sim(Object Name) $\geq$ 0.9 $\Rightarrow$ Title equal		103917	0.014
1	sim(Object Name) $\geq$ 0.9 $\Rightarrow$ Medium equal		103917	0.084
2	sim(Object Name) $\geq$ 0.9 $\Rightarrow$ Classification equal		103917	0.615
3	sim(Title) $\geq$ 0.9 $\Rightarrow$ Object Name equal		1639	0.948
4	sim(Title) $\geq$ 0.9 $\Rightarrow$ Medium equal		1639	0.184
5	sim(Title) $\geq$ 0.9 $\Rightarrow$ Classification equal		1639	0.277
6	sim(Medium) $\geq$ 0.9 $\Rightarrow$ Object Name equal		19195	0.476
7	sim(Medium) $\geq$ 0.9 $\Rightarrow$ Title equal		19195	0.011
8	sim(Medium) $\geq$ 0.9 $\Rightarrow$ Classification equal		19195	0.705
9	sim(Classification) $\geq$ 0.9 $\Rightarrow$ Object Name equal		82490	0.764
10	sim(Classification) $\geq$ 0.9 $\Rightarrow$ Title equal		82490	0.004
11	sim(Classification) $\geq$ 0.9 $\Rightarrow$ Medium equal		82490	0.152

Figure 5: Matching Dependency for object-related attributes

Out[]:		MD	support	confidence
0	sim(Region) $\geq$ 0.9 $\Rightarrow$ Country equal		1606	0.925
1	sim(Subregion) $\geq$ 0.9 $\Rightarrow$ Country equal		271	0.985
2	sim(Subregion) $\geq$ 0.9 $\Rightarrow$ Region equal		271	1.000

Figure 6: Matching Dependency for geographical attributes

Finally, Figure 7 shows artist-related attributes, highlighting strong bidirectional dependencies between **Artist Display Name** and **Artist Alpha Sort**, which supports their use in wrangling task.

Out[]:		MD	support	confidence
0	sim(Artist Display Name) $\geq$ 0.9 $\Rightarrow$ Artist Alpha ...		2012	0.984
1	sim(Artist Alpha Sort) $\geq$ 0.9 $\Rightarrow$ Artist Display ...		2013	0.984

Figure 7: Matching Dependency for artist-related attributes

Association Rule Association rule mining was applied to discover frequent co-occurrence patterns among semantically related attributes. The analysis was conducted separately on object-related attributes (e.g., **Object Name**, **Title**, **Medium**, **Classification**) and geographical attributes (e.g., **Subregion**, **Region**, **Country**), in order to extract meaningful and interpretable rules.

Each record was first transformed into a transaction composed of attribute-value pairs. This transformation enables the application of itemset mining algorithms on categorical

data. The *TransactionEncoder* was used to convert the transactional representation into a binary matrix, where each column corresponds to an item and each row to a transaction. Frequent itemsets were then extracted using two well-known algorithms: *FP-Growth*, which efficiently mines frequent patterns without explicit candidate generation, and *ECLAT*, which relies on a vertical data representation to compute itemset supports through set intersections. Based on the frequent itemsets, association rules were generated by imposing minimum thresholds on support and confidence and further analysed through visualisation. The complete implementation for association rule mining is available in the project. However, due to the large size of the dataset and limited hardware resources, the full execution of the code was not performed.

2.2 Data Quality Assessment

Cardinality Dimensions: The quality of the dataset was assessed using cardinality-based dimensions computed for each column. The results are reported in Table 2. In particular, the following measures were considered:

- **Uniqueness:**

$$\text{uniqueness}(col) = \frac{\text{Cardinality}(col)}{n_{\text{rows}}}$$

- **Distinctness:**

$$\text{distinctness}(col) = \frac{\text{Cardinality}(col)}{\# \text{notNaN}}$$

- **Constancy:**

$$\text{constancy}(col) = \frac{\text{Mode}(col)}{n_{\text{rows}}}$$

Completeness:

$$\text{completeness} = \frac{\# \text{notNaN cells}}{n_{\text{rows}} \times n_{\text{cols}}}$$

Before data processing, the completeness score was equal to **0.5070**, meaning that approximately half of the dataset entries contained non-missing values. This relatively low level of completeness highlights the presence of a substantial amount of missing data, which motivated the subsequent data cleaning and preprocessing activities.

Consistency: was evaluated by verifying a set of structural and semantic rules on each tuple of the dataset. The assessment focuses on string validity and temporal coherence across selected attributes.

First, all columns containing missing values and having a non-boolean data type were identified. For each tuple, a *string consistency check* was applied to categorical attributes, ensuring that non-missing values are not empty or whitespace-only strings. Each satisfied string check contributes a fixed score to the overall consistency sum. In addition, *date consistency checks* were performed to validate temporal coherence. Specifically, when present, the **Artist Begin Date** was required to be less than or equal to the **Artist End Date**, and similarly the **Object Begin Date** was required to precede or coincide with the **Object End Date**. Each satisfied temporal constraint contributed equally to the consistency score.

Let R denote the total number of consistency rules applied to each tuple and n_{tuples} the number of tuples in the dataset. The final consistency measure was computed as:

$$\text{Consistency} = \frac{\text{consistent_tuples_scores_sum}}{n_{\text{tuples}} \times R}$$

Before data processing, the consistency assessment yielded a score of **0.7360**, indicating a moderately high level of conformity of the raw dataset with respect to the defined structural and temporal consistency rules.

Accuracy: this dimension could not be evaluated, as no ground truth information was provided with the dataset.

Timeliness: this dimension could not be assessed, since no temporal information regarding data currency and volatility was available.

2.3 Data Wrangling

The data wrangling phase transformed the raw dataset into a consistent and usable format. This process followed the execution flow of the provided pipeline, iterating through attributes to clean, standardize, and optimize them step-by-step.

2.3.1 Initial Dimensionality Reduction

The first step of the data wrangling process involved reducing the dimensionality of the dataset by removing attributes that were either statistically insignificant due to missing data or semantically irrelevant to the analysis.

High Missing Value Threshold: An analysis of the missing value ratio per column was performed. A threshold of **95%** was established, meaning any attribute with less than **5%** valid data was considered insufficient for imputation or analysis. Based on this rule, the following 10 columns were dropped:

- Artist Suffix, Reign, State, Locale, Locus;
- Portfolio, River, County, Subregion, Excavation.

Removing these sparse columns significantly reduced the noise in the dataset without discarding valuable information.

Removal of Administrative Attributes: Several columns contained metadata solely for database management or museum administration. These attributes do not describe the artworks themselves and were therefore removed to streamline the dataset:

- Object Number and Object ID: Internal identifiers useful for database indexing but irrelevant for pattern mining.
- Metadata Date and Repository: Constant or administrative timestamps that provide no analytical value.
- Link Resource: URLs pointing to the museum’s website, which are not usable for statistical analysis.

Column Name	Uniqueness	Distinctness	Constancy
Object Number	0.9943	0.9943	0.0000
Is Highlight	0.0000	0.0000	0.9959
Is Public Domain	0.0000	0.0000	0.5489
Object ID	1.0000	1.0000	0.0000
Department	0.0000	0.0000	0.3446
Object Name	0.0580	0.0583	0.1987
Title	0.4983	0.5359	0.0158
Culture	0.0157	0.0376	0.0495
Period	0.0036	0.0227	0.0194
Dynasty	0.0007	0.0143	0.0152
Reign	0.0007	0.0303	0.0054
Portfolio	0.0063	0.1380	0.0013
Artist Role	0.0126	0.0218	0.2406
Artist Prefix	0.0114	0.0575	0.0629
Artist Display Name	0.1258	0.2159	0.0216
Artist Display Bio	0.0913	0.1827	0.0216
Artist Suffix	0.0036	0.1587	0.0053
Artist Alpha Sort	0.1258	0.2160	0.0216
Artist Nationality	0.0085	0.0194	0.1415
Artist Begin Date	0.0464	0.0966	0.0243
Artist End Date	0.0474	0.0997	0.0226
Object Date	0.0654	0.0678	0.0377
Object Begin Date	0.0046	0.0046	0.0526
Object End Date	0.0045	0.0045	0.0465
Medium	0.1310	0.1334	0.0461
Dimensions	0.5304	0.6169	0.0058
Credit Line	0.0808	0.0809	0.0931
Geography Type	0.0003	0.0021	0.0591
City	0.0055	0.0813	0.0101
State	0.0025	0.1324	0.0020
County	0.0002	0.0406	0.0034
Country	0.0023	0.0136	0.0690
Region	0.0015	0.0220	0.0248
Subregion	0.0008	0.0164	0.0068
Locale	0.0017	0.0517	0.0043
Locus	0.0029	0.1856	0.0009
Excavation	0.0007	0.0210	0.0053
River	0.0005	0.1079	0.0008
Classification	0.0024	0.0028	0.1545
Rights and Reproduction	0.0025	0.0492	0.0212
Link Resource	1.0000	1.0000	0.0000
Metadata Date	0.0000	0.0000	1.0000
Repository	0.0000	0.0000	1.0000

Table 2: Data quality assessment metrics per column

2.3.2 Data Type Optimization: Artist Dates

The temporal attributes **Artist Begin Date** and **Artist End Date** were originally interpreted by the pandas library as generic object types (strings). This format was unsuitable for analysis as it prevented numerical operations, such as calculating the artist’s lifespan or verifying temporal consistency.

To address this, a specific type conversion was applied:

- **Target Type:** The columns were cast to the `Int64` data type (a nullable integer type in pandas).
- **Handling Missing Values:** Unlike the standard numpy `int64`, the pandas `Int64` type supports NaN values. This allowed us to convert valid year strings into integers while preserving missing entries without being forced to use floating-point representations.

This optimization ensured that all present values were treated as valid years, facilitating the subsequent dependency checks and feature extraction steps.

2.3.3 Column-by-Column Standardization

The pipeline iterated through specific attributes to analyze their cardinality, apply text cleaning where necessary, and optimize data types. **Frequency distributions** were analyzed to verify the effectiveness of the cleaning operations.

Department: The **Department** column contained 20 unique values, representing the high-level curatorial divisions of the museum. The most frequent department was “Drawings and Prints” (154,445 occurrences). No text cleaning was required, and the column was cast to a `categorical` type to optimize memory usage.

Object Name: This attribute exhibited high cardinality (25,987 unique values). The most frequent values were “print” (89,052) and “photograph” (28,440). Although the vocabulary was diverse, the data was consistent enough to be cast directly to a `categorical` type without aggressive string manipulation.

Culture: The **Culture** column initially contained complex strings characterized by uncertainty markers (e.g., “possibly American”) and parenthetical qualifiers. A custom cleaning function was applied to standardize these values:

- **Regex Cleaning:** Text inside parentheses was removed, and uncertainty phrases such as “probably”, “possibly”, “style of”, and “for export” were stripped to retain only the core cultural designation.
- **Normalization:** Separators were standardized, and whitespace was trimmed.

After cleaning, the number of unique values was significantly reduced, allowing for a clearer distribution analysis. The top 20 most frequent cultures are shown in Figure 8.

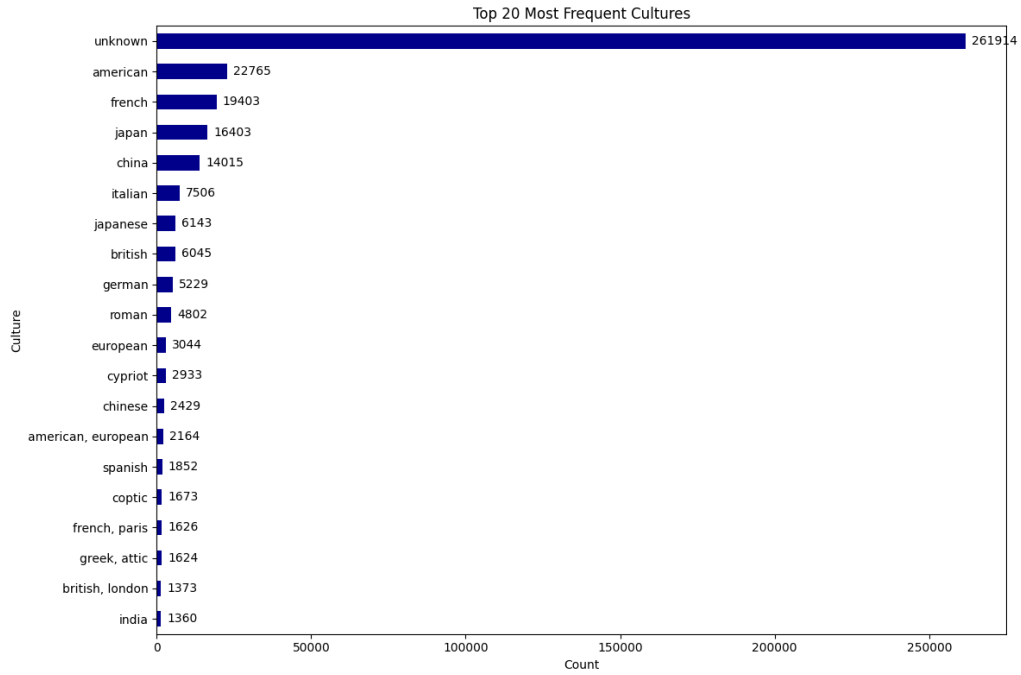


Figure 8: Top 20 most frequent cultures after cleaning

Artist Role: The **Artist Role** column (5,665 unique values) required standardization of separators, as multiple roles were often delimited by pipes or slashes. A cleaning function was applied to normalize these delimiters and remove extraneous whitespace. As shown in Figure 9, the generic role “Artist” is the most predominant, followed by specific roles such as “Publisher” and “Designer”.

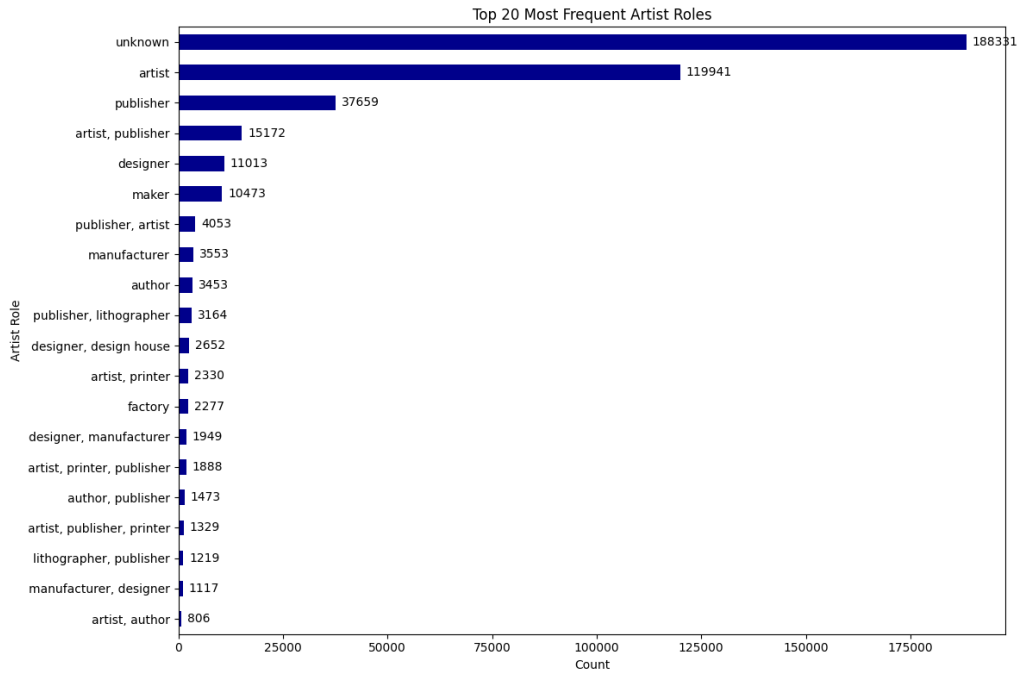


Figure 9: Top 20 most frequent artist roles

Artist Nationality: Similar to the culture attribute, **Artist Nationality** (3,798 unique values) contained uncertainty markers and combined nationalities. Regex-based cleaning was applied to remove qualifiers and standardize the text. The distribution of the top 20 nationalities is reported in Figure 10.

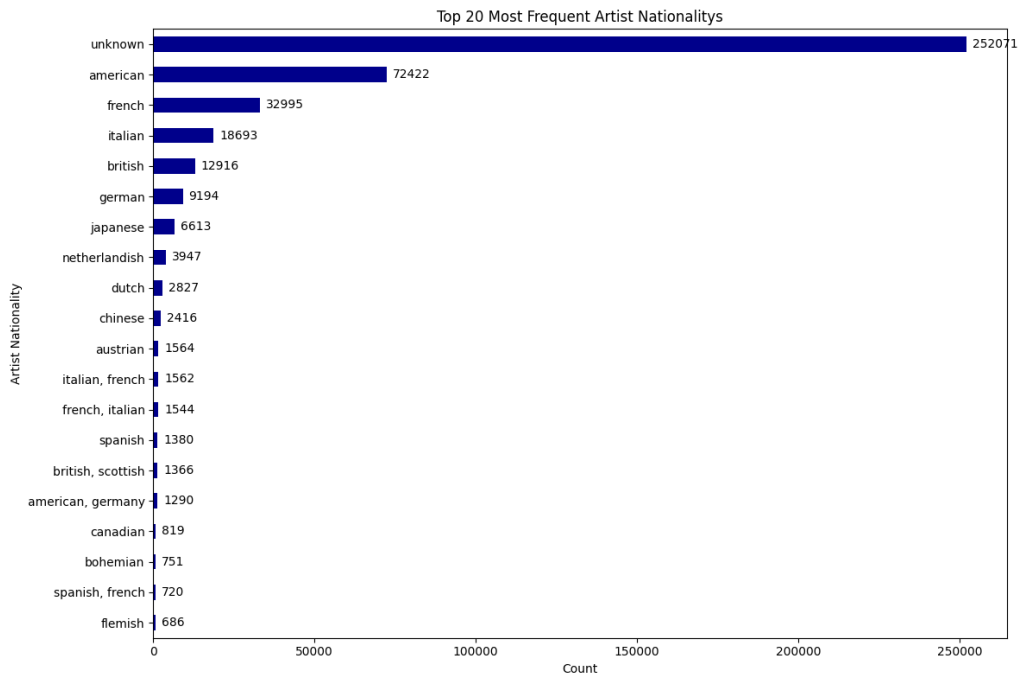


Figure 10: Top 20 most frequent artist nationalities

Medium: The **Medium** column had the highest cardinality among the categorical attributes (58,716 unique values) due to highly detailed descriptions of materials and techniques. The cleaning process involved:

- Splitting combined materials (e.g., “oil on canvas”).
- Removing parenthetical technical details.
- Standardizing variations of common terms (e.g., “gold”, “silver”).

The distribution of the most frequent mediums after cleaning is shown in Figure 11.

Department: The **Department** column contained 20 unique values, with “Drawings and Prints” being the most frequent (154,445 occurrences). No text cleaning was required. The column was cast to a categorical type to optimize memory usage.

Object Name: This attribute had high cardinality (25,987 unique values). The most frequent values were “print” (89,052) and “photograph” (28,440). While the variety of terms was high, the data was deemed consistent enough to be cast directly to a categorical type.

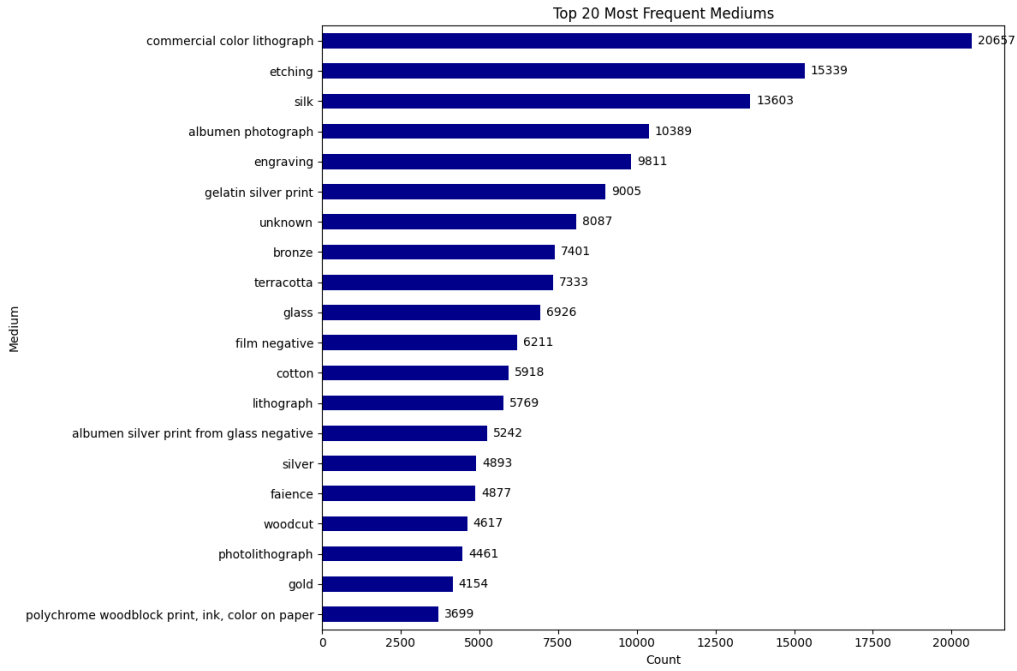


Figure 11: Top 20 most frequent mediums

Culture: The **Culture** column initially contained complex strings with parentheses and uncertainty markers (e.g., “possibly American”). A custom cleaning function was applied to:

- Remove text inside parentheses.
- Remove uncertainty phrases like “probably”, “possibly”, and “for export”.
- Normalize separators (e.g., changing “ or ” to commas).

After cleaning, the number of unique values was reduced to 4,800. The most frequent culture was “American” (22,765), excluding unknown values. The column was then cast to categorical.

Artist Role: The **Artist Role** column contained 5,665 unique values. A cleaning function was used to normalize separators (pipes, slashes) and remove whitespace. The most common role identified was “Artist” (119,941), followed by “Publisher” and “Designer”. The column was converted to categorical.

Artist Nationality: Similar to Culture, the **Artist Nationality** column (3,798 unique values) required regex-based cleaning to remove uncertainty markers and standardize separators. “American” was the most frequent nationality (72,422). The column was optimized as a categorical type.

Medium: The **Medium** column exhibited very high cardinality (58,716 unique values) due to detailed material descriptions. A cleaning function standardized the text by splitting combined materials and removing parenthetical details. The most frequent medium was “commercial color lithograph” (20,657). It was cast to categorical.

2.3.4 Secondary Dimensionality Reduction

Following the standardization of key attributes, a second dimensionality reduction phase was executed. This step aimed to eliminate features that, despite passing the initial missing value threshold, provided low information gain or contained redundant data.

Redundant and Low-Information Columns: The following attributes were dropped due to their limited utility for pattern discovery:

- **Geographical Redundancy:** `Region` and `Geography Type` were removed as they offered less specificity compared to `Country` and `City`, or contained vague classifications.
- **Unstructured Text:** `Credit Line`, `Dimensions`, and `Rights and Reproduction` contained highly variable, free-text descriptions that were difficult to standardize without advanced natural language processing.
- **Sparse Historical Data:** `Dynasty` and `Artist Prefix` were dropped because they applied to only a small fraction of the dataset, creating excessive sparsity.

Attribute Selection:

- **Artist Name Selection:** The dataset contained both `Artist Display Name` and `Artist Alpha Sort`. To avoid duplication, `Artist Display Name` was dropped in favor of `Artist Alpha Sort`, which provides a normalized format (Last Name, First Name) more suitable for sorting and grouping.
- **Period Removal:** The `Period` column was discarded. Although potentially useful, it exhibited a high volume of missing values and lacked strong correlation with other verified temporal features like `Object Date`, making it unreliable for consistent analysis.

2.3.5 Classification Optimization

The `Classification` column serves as a high-level taxonomy for the collection, categorizing objects into groups such as “Prints”, “Photographs”, and “Textiles”. Analysis confirmed the presence of 1,078 unique categories, with “Prints” being the most frequent classification. Given the structural consistency of this attribute, it was converted to a `categorical` data type. This transformation significantly reduced the memory footprint and prepared the feature for efficient grouping and analysis.

2.3.6 Feature Extraction: Temporal Attributes

A significant challenge in the dataset was the presence of missing values in the structured temporal columns (`Artist Begin/End Date` and `Object Begin/End Date`), despite distinct year information being available in the unstructured text descriptions. To address this, a feature extraction strategy based on Regular Expressions (regex) was implemented.

Artist Bio Extraction: The `Artist Display Bio` column often contained life dates in natural language formats (e.g., “born 1850”, “1850–1910”). A custom function was applied to:

- Identify 4-digit year patterns within the text.
- Assign the first identified year to `Artist Begin Date` and the second to `Artist End Date` if the structured columns were missing (NaN).

This operation significantly reduced the null count for artist lifespan attributes.

Object Date Extraction: Similarly, the `Object Date` column provided unstructured descriptions of creation times (e.g., “ca. 1800”, “19th century”). A regex pattern was used to extract the first valid year sequence found. This value was used to impute missing entries in `Object Begin Date` and `Object End Date`, ensuring a more complete temporal dimension for the analysis.

Final Cleanup: Following the successful extraction and imputation of these dates, the original source columns `Artist Display Bio` and `Object Date` were removed. This eliminated redundant textual data and ensured that all temporal information was stored strictly in numerical format.

2.4 Inconsistency Detection

Following the structural formatting and type conversion, a comprehensive consistency assessment was performed. This step aimed to differentiate between data that is syntactically correct (valid types) and data that is semantically correct (valid logic).

Temporal Consistency Checks: Three specific constraints were applied to validate the coherence of date-related attributes.

- **Artist Lifespan:** A logical check ensured that an artist’s birth year must precede or equal their death year:

$$\text{Artist Begin Date} \leq \text{Artist End Date}$$

- **Object Creation Period:** A similar validity check was applied to the artwork’s timeline, ensuring the start date does not follow the end date:

$$\text{Object Begin Date} \leq \text{Object End Date}$$

- **Historical Coherence:** A cross-column validation was implemented to ensure historical logic, specifically that the artist was born before the artwork was created (allowing for a small tolerance margin for attribution ambiguity):

$$\text{Artist Begin Date} \leq \text{Object Begin Date}$$

Semantic Consistency and Mismatch Detection: Beyond temporal rules, semantic checks were conducted to verify the alignment of textual attributes.

- **Title vs. Object Name:** A semantic comparison was performed between the **Title** and **Object Name** columns. Instances where the specific description in the **Title** (e.g., “Bowl with...”) contradicted the category in **Object Name** (e.g., “Coin”) were flagged.
- **Value Mismatches:** A generalized check identified tuples where related attributes held contradictory information, such as a **City** not belonging to the specified **Country**, or a **Medium** description (e.g., “Oil on canvas”) conflicting with the **Classification** (e.g., “Textiles”).

Outcome of Inconsistency Analysis: This diagnostic phase revealed that despite successful type conversion, the dataset contained multiple logical violations. Specifically, several records exhibited negative lifespans (birth year > death year), likely due to data entry typos or column swapping. Additionally, historical inconsistencies were identified where artwork dates predated the artist’s birth, highlighting attribution errors. These findings defined the specific cleaning rules applied in the subsequent phase.

2.5 Missing Value Handling

Following the inconsistency checks, the pipeline addressed the remaining missing values. As shown in the data profiling stage, several attributes exhibited high sparsity. The imputation strategy was designed to preserve the statistical integrity of the dataset without introducing artificial bias.

2.5.1 Numerical Imputation (Temporal Data)

Following the regex-based feature extraction, missing values remained in the **Artist** temporal columns. To ensure data integrity, records lacking an **Artist Begin Date** or **Artist End Date** were removed from the dataset.

2.5.2 Categorical Imputation

For categorical attributes, a distinction was made between features with high missingness and those with low missingness.

High Sparsity Attributes (Culture, Nationality, Role): Columns such as **Culture** (approx. 58% missing), **Artist Nationality**, and **Artist Role** contained too many missing values to be reliably imputed using statistical modes or predictive models without distorting the data distribution.

- **Strategy:** A placeholder token “**Unknown**” was assigned to these missing entries.
- **Rationale:** This approach explicitly models the absence of information as a distinct category, preventing the algorithm from learning false associations based on guessed values.

Low Sparsity Attributes (Medium, Classification): Attributes with lower missing rates, such as `Classification` and `Medium`, were imputed using the **Mode** (most frequent value) of the respective column. This method assumes that the missing entries are likely to belong to the dominant category in the collection (e.g., assigning “Prints” to uncategorised artworks).

Textual Attributes (Title): Missing entries in the `Title` column were filled with the standard value “`Untitled`”. This ensured that text-based operations (such as duplicate detection) could proceed without errors caused by null values.

2.5.3 Final Cleaned Dataset State

After these imputation steps, the dataset was fully dense (no `NaN` values). This complete dataset served as the input for the subsequent outlier detection and duplicate identification phases.

2.6 Outlier Handling

Outlier detection was a critical step in ensuring the historical accuracy of the Metropolitan Museum of Art dataset. Given the heterogeneous nature of the collection spanning ancient artifacts to art distinguishing between legitimate extreme values (e.g., objects from 2500 BC) and data errors (e.g., typos like year 2050) was a significant challenge. We adopted a multi-faceted approach combining statistical, distance-based, and domain-logic methods.

2.6.1 Statistical Methods (Univariate Analysis)

We applied univariate statistical techniques to identify anomalies in continuous temporal variables: `Object Begin Date`, `Object End Date`, and the derived `Artist Lifespan`.

Object Temporal Data We compared three statistical methods to determine the most robust approach for historical dates. The results highlighted the sensitivity of parametric methods to the “masking effect” in heavy-tailed historical data:

- **Z-Score (Parametric):** Assumes a Gaussian distribution. It identified approximately 600–680 outliers per column. This method was too conservative, failing to flag significant deviations due to the inflated standard deviation caused by extreme values.
- **IQR (Interquartile Range):** A robust method that identified over 5,200 outliers in the `Object End Date`. While effective, the outlier range (-2575 to 1531) was too narrow, flagging legitimate ancient artifacts (e.g., Egyptian antiquities) as errors.
- **MAD (Median Absolute Deviation):** Identified $\sim 3,350$ outliers in the `End Date` column. This served as a middle ground but still required domain interpretation.

Decision: We utilized the IQR range for initial visualization but relied on **Domain Logic** for the final cleaning to preserve ancient history (e.g., years < -2000).

Artist Lifespan Analysis We calculated a derived feature, **Lifespan** (**Artist End** – **Artist Begin**), to detect anomalies. The analysis revealed a clear distinction between human artists and corporate entities.

- **Mean Lifespan:** 73.9 years ($\sigma = 29.9$).
- **Z-Score Threshold:** Flagged lifespans > 163.7 years ($Z > 3$), identifying 2,676 outliers.
- **MAD Threshold:** Flagged lifespans > 144 years, identifying 3,667 outliers.

Upon inspection, we discovered that records exceeding the MAD threshold were not typos but **Institutions** (e.g., factories, companies) misclassified as individuals. Table 3 illustrates these findings.

Artist Name	Begin	End	Lifespan	Entity Type
William, Adams & Sons	1814	2063	249	Institution
Rose Factory	1662	1854	192	Institution
Libbey Glass Company	1888	2050	162	Institution
Noritake — Wright, F. L.	1904	2050	146	Collaboration

Table 3: Sample of anomalies detected via Statistical Lifespan Analysis. These entries represent institutions with "dynastic" lifespans rather than biological errors.

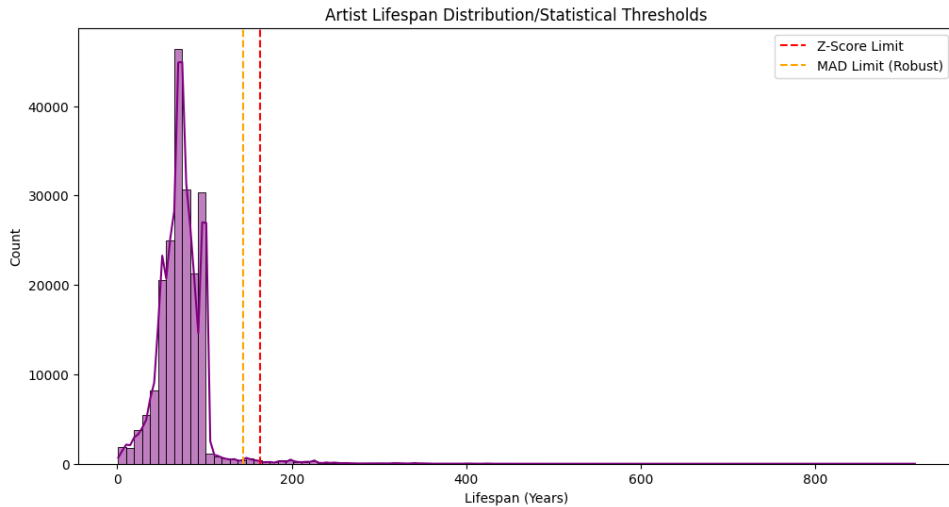


Figure 12: Distribution of Artist Lifespans. The vertical lines indicate the Z-Score and MAD thresholds. The long tail to the right indicates the presence of non-human entities.

2.6.2 Distance-Based Methods (Multivariate Analysis)

To capture anomalies in the *relationship* between dates, we employed the **Local Outlier Factor (LOF)** algorithm.

- **Rationale:** LOF detects samples with substantially lower density than their neighbors. This is effective for identifying objects where the span between **Begin** and **End** is unusual, even if the individual years are valid.

- **Input:** 2D feature space of [Object Begin Date, Object End Date].
- **Results:** The LOF algorithm identified **2,118** outliers. These points highlighted structural anomalies, including a severe data entry error where a year was recorded as **157,555**.

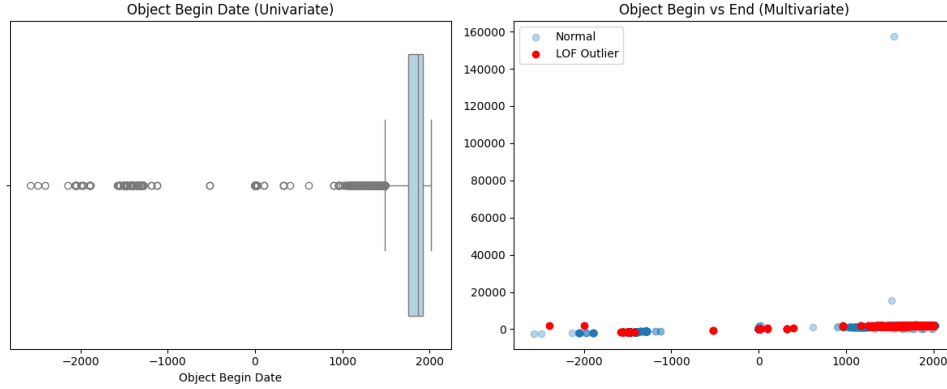


Figure 13: Local Outlier Factor (LOF) Analysis. Red points indicate density-based outliers, successfully isolating extreme data entry errors and impossible time ranges.

2.6.3 Model-Based Logic and Final Cleaning

Instead of a *black-box* model, we implemented a **Logic-Based Validity Model** to finalize the cleaning. This approach enforces strict causal and domain constraints.

1. **Causality Constraint:** $\text{Object End} \geq \text{Object Begin}$. Our final logic check confirmed 0 records remained with impossible negative durations.
2. **Future Constraint:** Dates cannot exceed the current era. We capped the max year at 2025, correcting typos such as "2050" or "2063" found in the raw data.
3. **Entity Classification Model:** Using the MAD threshold (> 144 years), we split the **Artist** entity into two distinct categories rather than deleting data.

Final Outcome The cleaning process yielded a dataset of **203,488** verified records.

- **Temporal Range:** Corrected to $[-2575, 2024]$, preserving valid ancient history while removing typos.
- **Artist Classification:** 200,558 identified as *Persons* and 2,930 reclassified as *Institutions*.

2.7 Deduplication

Duplicate detection is essential to prevent statistical bias and ensure that the museum's inventory is not artificially inflated. Given the inconsistencies in historical data entry (e.g., "New York" vs. "New York City"), exact string matching was insufficient. We implemented a **Fuzzy Deduplication** pipeline using the `recordlinkage` library.

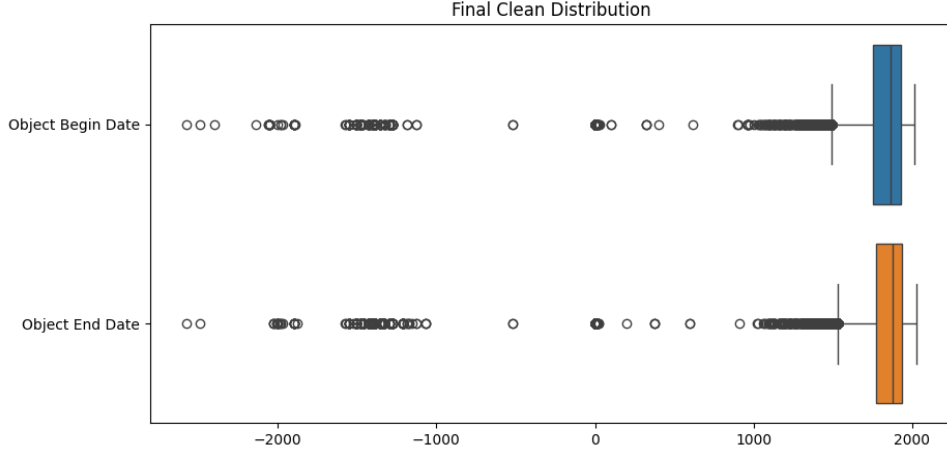


Figure 14: Final Clean Distribution. The extreme outliers (e.g., year 157,555) have been removed, while the valid "long tail" of ancient history (left whiskers) is preserved.

2.7.1 Methodology

To efficiently handle the large dataset, we employed a **blocking strategy** to reduce the computational cost of comparing every record against every other record ($O(N^2)$).

1. **Blocking Key:** We created a "signature" for each record based on the **Artist Alpha Sort** and the first 3 characters of the **Object Name**. This reduced the search space by only comparing records that shared similar primary attributes.
2. **Comparison Metrics:**
 - **Artist Name:** Jaro-Winkler similarity (Threshold > 0.90).
 - **Object Name:** Jaro-Winkler similarity (Threshold > 0.85).
 - **City:** Levenshtein distance (Threshold > 0.80).
 - **Date Logic:** Exact match required for **Object Begin Date** (to distinguish different works by the same artist).

2.7.2 Results

The deduplication process yielded the following statistics:

- **Unique Signatures Processed:** 65,500
- **Candidate Pairs Generated:** 293,750
- **Potential Duplicates Identified:** 18,845

Table 4 presents a sample of the identified pairs. The "Grouped" view highlights that while the metadata is nearly identical, the records possess distinct IDs.

Pair ID	Record ID	Artist Alpha Sort	Object Name	City
2*1	1145	United States Pottery Company	Frame	Bennington
	1144	United States Pottery Company	Frame	Bennington
2*2	56272	Wilkinson, Charles K.	Facsimile, Aashyt, Sarcophagus	Unknown
	56270	Wilkinson, Charles K.	Facsimile, Aashyt, Sarcophagus	Unknown
2*3	2946	Bierstadt, Albert	Painting	Unknown
	2181	Bierstadt, Albert	Painting	Unknown
2*4	2904	Svinin, Pavel P. — Krimmel, John L.	Watercolor	Unknown
	2902	Svinin, Pavel P. — Krimmel, John L.	Watercolor	Unknown
2*5	2888	Stuart, Gilbert	Painting	Unknown
	813	Stuart, Gilbert	Painting	Unknown

Table 4: Sample of Fuzzy Duplicates Detected. Each block represents a pair of records identified as potential duplicates. The identical metadata values across distinct Record IDs suggest redundant entries or multiple copies of the same work.

2.7.3 Resolution Strategy

Given the museum context, we rejected the naive approach of deleting "duplicate" records, as identical metadata often refers to physically distinct objects (e.g., multiple prints of the same lithograph). Instead, we implemented a **Metadata Standardization** strategy to correct inconsistencies without data loss.

- **Record Preservation:** No rows were removed from the dataset. All 18,845 potential duplicates were retained to ensure inventory completeness.
- **Value Unification:** We utilized the Record Linkage results to generate a mapping of "variant" spellings to "canonical" forms. The values for *Artist Alpha Sort* and *Object Name* were standardized by propagating the text from the reference record ('idx_good') to the matched record ('idx_bad').
- **Conditional Geography Update:** To avoid false positives in location data, *City* names were only standardized if the specific field similarity score exceeded 0.80.

3 Results

Following the extensive cleaning, imputation, and deduplication processes, we re-evaluated the dataset to quantify the improvement in data quality. This final assessment mirrors the methodology used in the initial Data Profiling phase.

3.1 Dimension Definitions

To ensure objective measurement, we utilized the following standard quality dimensions:

- **Completeness:** The ratio of non-missing values to the total number of records. A completeness of 1.0 indicates a fully dense dataset.
- **Consistency:** The degree to which data values conform to a specific format or domain rule (e.g., valid date ranges, correct data types).
- **Uniqueness:** The ratio of distinct values (cardinality) to the total number of records. High uniqueness is expected for identifiers (e.g., IDs) and textual content (e.g., Titles), while low uniqueness is expected for categorical fields.

- **Constancy:** The frequency of the most common value (mode) in a column. A very high constancy (near 1.0) suggests a column with little variation (e.g., a boolean flag that is almost always "False").

3.2 Dimension Measurements

The final quality assessment confirms that the pipeline successfully addressed the initial data deficits.

3.2.1 Structural Analysis

Table 5 presents the structural properties of the cleaned dataset. notably, the **Completeness** score reached **1.0** (100%) across all columns, as all missing values were successfully imputed or handled. The dataset now contains **337,142** valid records.

Attribute	Total Records	Missing	Cardinality
Is Highlight	337,142	0	2
Is Public Domain	337,142	0	2
Department	337,142	0	19
Object Name	337,142	0	3,895
Title	337,142	0	142,980
Culture	337,142	0	916
Artist Role	337,142	0	3,003
Artist Alpha Sort	337,142	0	42,822
Artist Nationality	337,142	0	1,473
Object Begin Date	337,142	0	785
Object End Date	337,142	0	770
Medium	337,142	0	25,175
Classification	337,142	0	608
City	337,142	0	966
Country	337,142	0	116

Table 5: Final Structural Analysis. The dataset is now fully complete (0 missing values), providing a solid foundation for analysis.

3.2.2 Content Quality Assessment

Table 6 details the content distribution metrics.

- **Consistency:** The overall consistency score of the dataset was calculated at **0.9894**. This high score reflects the successful standardization of categorical values and the logical validation of date fields.
- **Constancy & Uniqueness:** The metrics show expected behavior for a curated dataset. **Title** has high uniqueness (0.42), reflecting the diversity of the collection, while administrative flags like **Is Highlight** show high constancy (0.99).

Attribute	Uniqueness	Constancy
Is Highlight	0.0000	0.9963
Is Public Domain	0.0000	0.6278
Department	0.0001	0.5831
Object Name	0.0116	0.3762
Title	0.4241	0.0205
Culture	0.0027	0.8328
Artist Role	0.0089	0.4874
Artist Alpha Sort	0.1270	0.0674
Artist Nationality	0.0044	0.2579

Table 6: Final Content Quality Metrics. High constancy in administrative fields and healthy uniqueness in descriptive fields indicate a well-structured dataset.

4 Performance Analysis

To validate the effectiveness of our data quality pipeline beyond simple summary statistics, we performed a comparative analysis using dimensionality reduction and clustering metrics. This step assesses whether the cleaning process improved the *structural integrity* and *separability* of the data, making it suitable for downstream Machine Learning tasks.

4.1 Methodology

We compared the **Raw Dataset** (with missing values dropped to enable computation) against the **Clean Dataset** (imputed and standardized). The analysis followed these steps:

1. **Data Preparation:**

- **Encoding:** Categorical variables (e.g., **Department**, **Medium**) were transformed using Label Encoding.
- **Imputation Impact:** The Raw analysis required dropping rows with NaN values, reducing the dataset to **152,054** records. The Clean pipeline retained all **337,142** records through imputation.
- **Scaling:** Both datasets were normalized using **StandardScaler** to mean=0 and variance=1.

2. **Dimensionality Reduction (PCA):** We applied Principal Component Analysis (PCA) to reduce the 15-dimensional feature space to 12 components. This allowed us to measure how much variance was explained by the principal components.

3. **Manifold Learning (t-SNE):** To visualize the high-dimensional structure, we used t-Distributed Stochastic Neighbor Embedding (t-SNE). Unlike PCA, t-SNE preserves local neighborhood structures, making it ideal for visualizing clusters.

4.2 Quantitative Evaluation

The metrics in Table 7 demonstrate a substantial improvement in data quality.

Metric	Raw Data	Clean Data	Change
Sample Size	152,054	337,142	+121% (Retention)
Silhouette Score	0.2466	0.5425	+120% (Separability)
Explained Variance (PC1)	0.4805	0.0988	-79% (Noise Removal)

Table 7: Performance Metrics Comparison. The cleaning process improved separability (Silhouette) and removed artificial variance.

4.2.1 Interpreting the Variance Drop

A naive reading of PCA might suggest that higher explained variance (0.48) in the Raw data is better. However, our domain analysis reveals the opposite:

- **Raw Data (High Variance is Bad):** The 48% variance in the first component was driven by **Binary Noise**. The raw data was split heavily between "Complete Records" and "Broken Records" (or extreme outliers like year 157,555). The PCA simply picked up on this massive, artificial gap.
- **Clean Data (Low Variance is Good):** The drop to roughly 10% indicates that the "artificial" dominant factor has been removed. The variance is now spread more evenly across meaningful features (Time, Department, Culture), reflecting the true, complex nature of the art collection rather than data errors.

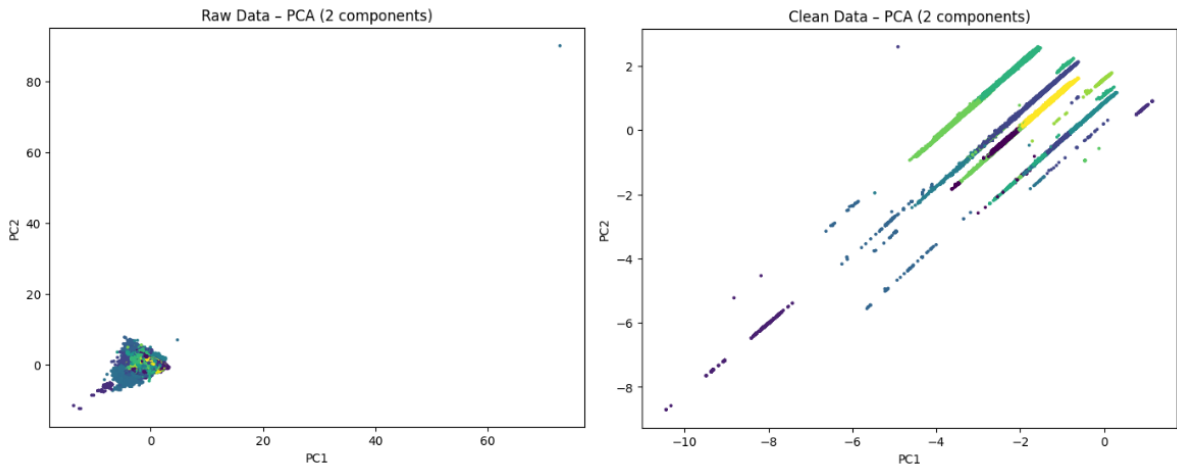


Figure 15: PCA Projection Comparison (PC1 vs PC2). **Left (Raw):** The plot is dominated by extreme outliers (the "long lines" stretching out), creating artificially high variance (48%) but low informational density. **Right (Clean):** The outliers are handled, resulting in a compact, dense cloud. The lower variance (10%) reflects the removal of these artificial distortions, adhering that valid data is often more compact than erroneous data.

4.3 Visual Evaluation (t-SNE)

The t-SNE visualization (Figure 16) visually confirms the quantitative metrics.

- **Raw Data (Left):** The plot shows a "blob-like" structure with weak separation. Classes overlap significantly, and the high variance suggests the presence of dominant artifacts (likely missing value patterns).
- **Clean Data (Right):** The plot reveals distinct, well-separated islands (clusters). This indicates that the cleaning process successfully disentangled the data, allowing the underlying relationships to emerge clearly.

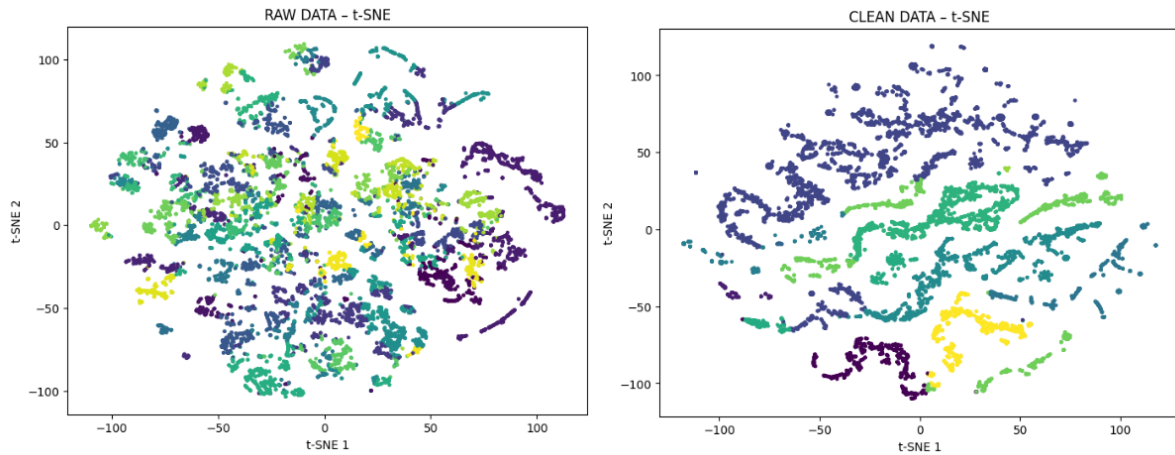


Figure 16: t-SNE Projection Comparison. The Clean Data (right) shows tighter, more distinct clusters compared to the diffuse structure of the Raw Data (left), indicating successful standardization and error removal.